

Midterm

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2026-05-14

Set Up

```
# make object halibut and read
halibut <- read.csv("halibut.csv")
# make object bee_pathogen and read
bee_pathogens <- read.csv("bee-pathogens.csv")
#read in tidyverse
library(tidyverse)
# read in ggplot2
library(ggplot2)
# read in janitor
library(janitor)
# read in flextable
library(flextable)
# read in rstatix
library(rstatix)
```

Problem 1. Understanding, critiquing, and adding to written communication

a. Test Type

The individual running this analysis could have run a two-sample t-test or a Mann-Whitney U Test. These tests make the most sense for the data set because the data is broken up into two groups based on the region of halibut, north and south. Since we don't know the variances we can not specify if it would be a Students or Welch's t-test. Additionally, in order to run a Mann-Whitney U Test there must be two groups, similar to a two sample t-test, so that is why

these tests work the best for the analysis. These tests are similar in the sense that they both require two different samples to be able to run. Additionally, they both compare a measurement of central tendencies. The major differences are two Sample T Tests are for groups with normal distributions or that meet the Central Limit Theorem, while Mann-Whitney U has to have random sampling of groups and the distributions just have to be similar in shape. Finally, while t tests usually compare means, Mann-Whitney U Tests compare medians.

b. Writing New Statements

Two-Sample T-Test: We found a moderate (Cohen's d) effect of regional location in California on halibut landing (lbs) when comparing northern halibut landings (n , $\bar{x}$) and southern (n , $\bar{x}$) halibut landing (Welch's or Student's two-sample t-test, $t(df)$, p-value, α).

Mann-Whitney U Test: We found a moderate (Cliff's δ = Cliff's delta value) difference in median halibut landings (lbs) between Northern California halibut landings ($\tilde{x}$ = sample median, n = sample size) and Southern ($\tilde{x}$ = sample median, n = sample size) California halibut landings (Wilcoxon rank-sum test, W = w-value, $p = 0.3$, α = significance level).

c. New Figures

```
#create new object from halibut
halibut_clean <- halibut |>
  #convert landings lb to hundred thousands
  mutate(landings_lb_hundred_thousands = landings_lb/100000)
```

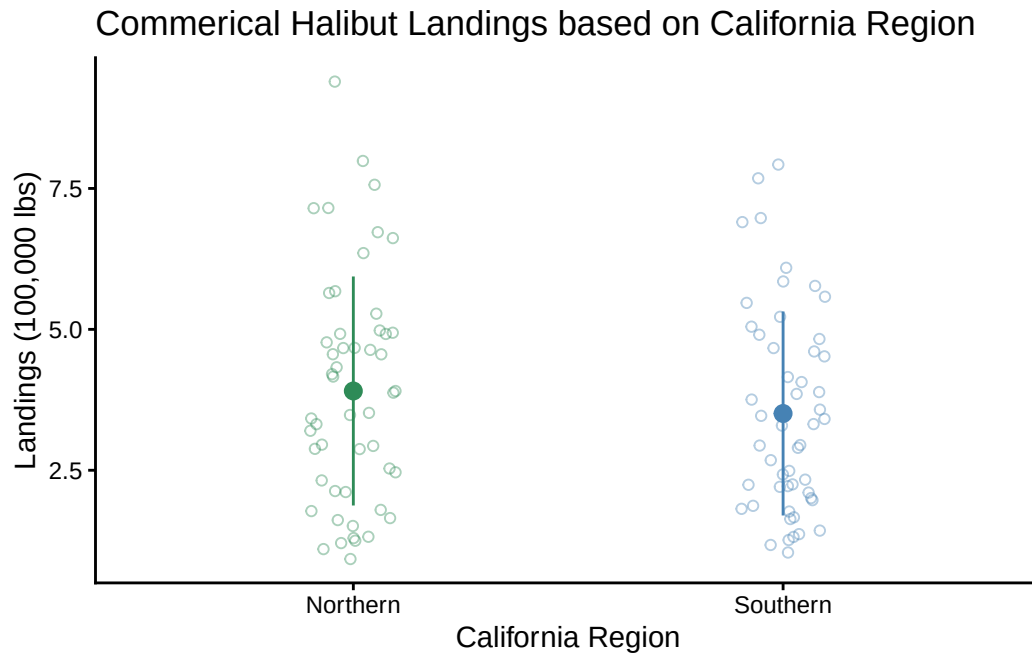
```
# store halibut as halibut_summary
halibut_summary <- halibut_clean |>
  # group the two treatments
  group_by(ca_region) |>
  # start the summary for halibut_summary data
  summarize(
    # calculate the mean weight
    mean_landings = mean(landings_lb_hundred_thousands),
    # calculate the number of observations
    n = n(),
    # calculate the standard deviation of weight
    sd_landings = sd(landings_lb_hundred_thousands))
# display summary calculations
halibut_summary
```

```
# A tibble: 2 x 4
  ca_region mean_landings      n sd_landings
  <chr>          <dbl> <int>      <dbl>
```

1 Northern	3.91	51	2.03
2 Southern	3.51	51	1.81

Two-Sample T-Test Figure:

```
# base layer: ggplot
ggplot(data = halibut_clean,
       # label x-axis
       mapping = aes(x = ca_region,
                     # label y-axis
                     y = landings_lb_hundred_thousands,
                     # set colors to treatment type
                     color = ca_region)) +
# first layer: adding data (each point shows an observation)
geom_jitter(width = 0.1,
            height = 0,
            alpha = 0.4,
            shape = 21) +
#second layer: means and standard deviation
geom_pointrange(data = halibut_summary,
               # label x- and y-values
               mapping = aes(x = ca_region,
                             y = mean_landings,
                             # set standard deviation below and above mean
                             ymin = mean_landings - sd_landings,
                             ymax = mean_landings + sd_landings)) +
# manually change colors
scale_color_manual(values = c("Northern" = "seagreen",
                              "Southern" = "steelblue"))+
# relabelling x- and y-axes, adding title
labs(x = "California Region",
     y = "Landings (100,000 lbs)",
     title = "Commerical Halibut Landings based on California Region") +
# set theme to minimal
theme_classic() +
# remove legend
theme(legend.position = "none")
```



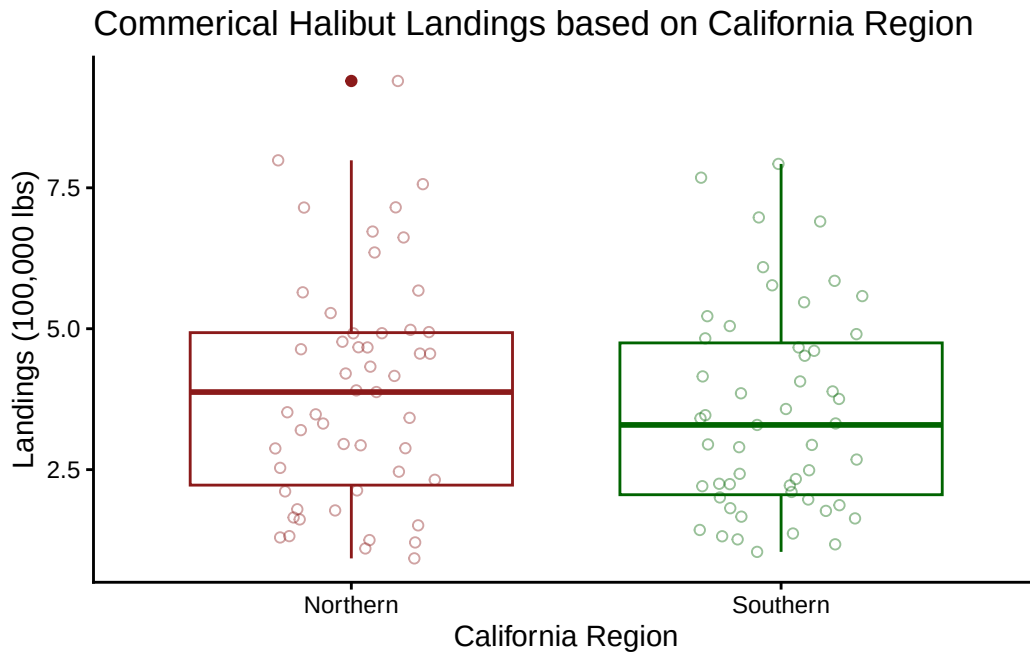
Mann-Whitney U Test Figure:

```
# base layer: ggplot
ggplot(data = halibut_clean,
       # x-axis: ca_region,
       # y-axis: landings_lb_hundred_thousands
       # color geometries by region
       mapping = aes(x = ca_region,
                     y = landings_lb_hundred_thousands,
                     color = ca_region)) +
# first layer: boxplot representing medians
geom_boxplot() +
# second layer: jitter representing individual ramen types
geom_jitter(height = 0,
            width = 0.2,
            alpha = 0.4,
            shape = 21) +
# using custom colors
scale_color_manual(values = c(
  "Northern" = "firebrick4",
  "Southern" = "darkgreen")) +
# relabelling x- and y-axes, adding title
labs(x = "California Region",
```

```

y = "Landings (100,000 lbs)",
title = "Commerical Halibut Landings based on California Region") +
# using a theme that's different from default
theme_classic() +
# taking out the legend
theme(legend.position = "none")

```



d. Caption Writing

Figure 1. Mean Halibut Landing Weights based on Region within California. Points represent measurements of halibut landings in hundred thousand pound from commercial fishing based in two regions of California (Green: Northern, Blue: Southern). Large darker dots represent the calculated mean landing weights measuring in hundred thousand pounds. The lines represent the calculated standard deviation measured in hundred thousand pounds. Data originally from: California Department of Fish and Wildlife. 2022. California halibut, *Paralichthys californicus*, Enhanced Status Report. Data accessed 2026-05-04.

Figure 2. Median Halibut Landing Weight for Regions in California. Points represent individual measurements of halibut landing weights in hundred thousand pounds from commercial fishing based in two regions of California (Red: Northern, Green: Southern). The bottom line of the box represent the 25th percentile. The middle line of the boxplot represents the median halibut landing weights, in the hundred thousand pounds, from each region. The top line of the boxplot represents the 75th percentile. The extending lines represent the further

point from 1.5 times the IQR. The filled in data point represents outliers outside of 1.5 times the IQR. Data originally from: California Department of Fish and Wildlife. 2022. California halibut, *Paralichthys californicus*, Enhanced Status Report. Data accessed 2026-05-04.

Problem 2. Data Visualization

a. Plan your Figure

- x-axis: column titled “Year”
- y-axis: column titled “Halibut Landings (LB)”
- geometry function: `geom_line()`
- colors and linetypes: column titled “ca_region”
- relabelling the x- and y-axes: `labs(x =, y =)`
- custom colors: `scale_color_manual()`
- Positioning the x-axis text at a 45 degree angle: `theme(axis.text.x = element_text(angle = 45))`
- putting the legend in the top right of the plot: `theme(legend.position = “top”, legend.justification = “right”)`
- removing the panel background: `theme(panel.background = element_blank())`
- removing the panel grid: `theme(panel.grid = element_blank())`
- removing the axis ticks: `theme(axis.ticks = element_blank())`
- removing the legend title: `theme(legend.title = element_blank())`

b. Code your Figure

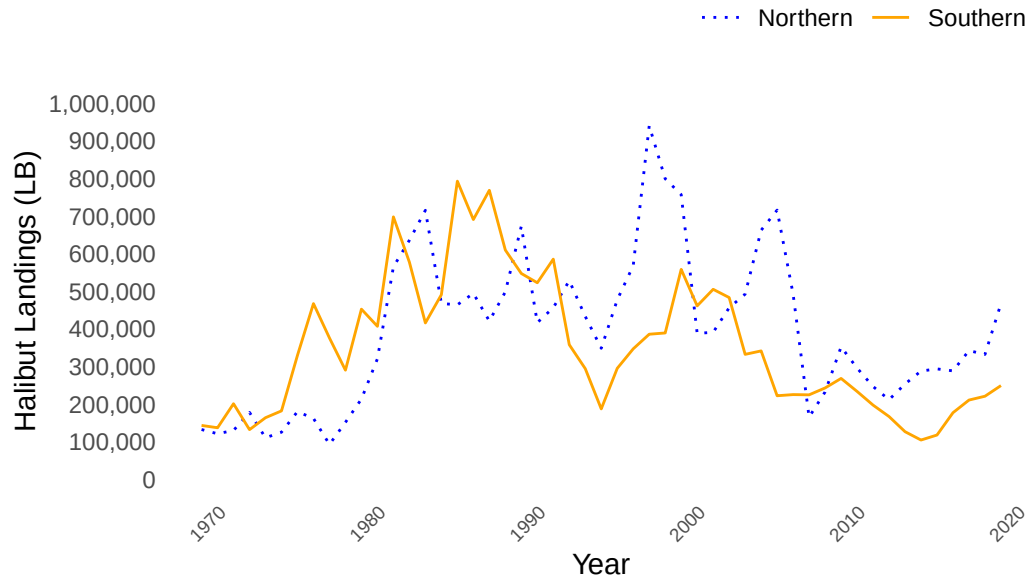
```
# base layer: ggplot
ggplot(data = halibut,
      # assign x and y values
      mapping = aes(x = year,
                    y = landings_lb,
                    # color by region
                    color = ca_region,
                    # line type by region
                    linetype = ca_region)) +
# first layer: geom_line
geom_line() +
```

```

# label x- and y-axes
labs(x = "Year",
     y = "Halibut Landings (LB)",
     # add subtitle with name
     subtitle = "Stephen Bond") +
# set manual colors for lines based on region
scale_color_manual(values = c("Northern" = "blue",
                              "Southern" = "orange")) +
# set manual line types based on region
scale_linetype_manual(values = c("Northern" = "dotted",
                                "Southern" = "solid")) +
# make y-axis ticks increase by 100,000
scale_y_continuous(breaks = seq(0, 1000000, by = 100000),
                  limits = c(0, 1000000),
                  labels = scales::comma) +
# tilt x-axis years by 45 degrees
theme(axis.text.x = element_text( angle = 45, hjust = 1, size = 7)) +
# place legend in top right
theme(legend.position = "top",
legend.justification = "right") +
#remove background
theme(panel.background = element_blank()) +
# remove grid line from panel
theme(panel.grid = element_blank()) +
# remove tick marks on the x-axis
theme(axis.ticks = element_blank()) +
# remove legend title
theme(legend.title = element_blank())

```

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c. Writing a Caption

Figure 3. Commercial Halibut Landings over time broken into Northern and Southern California Regions. The blue dashed line represents the data on halibut landings, measured in pounds, acquired from commercial fishing practices within northern California. The solid orange line represents data on halibut landings, measured in pounds, from commercial fishing practices within the southern California region. Data originally from: California Department of Fish and Wildlife. 2022. California halibut, *Paralichthys californicus*, Enhanced Status Report. Data accessed 2026-05-04.

Problem 3. Reproducing an analysis

a. Variables

- The response variable is pathogen load, specifically Deformed Wing Virus, Black Queen Cell Virus, *Vairimorpha bombi*, and *Crithidia bombi* measured in *Bombus impatiens* worker. This variable would be continuous.
- The predictor variable was the habitat type and the landscape variables. The groups for the habitat type were valley, edge, and forest, where as, the landscape variables ranged from non-forest natural, grassland/pasture, Developed open, Developed, Barren, Open water, crops, and to forest.

- The response variables were measured by performing quantitative PCRs on pooled abdomens of three *B. impatiens* workers per sample and analyzing 1-3 pools per site. These measurements then gave clear data on pathogen load within the sample (Gratton, E., et al., 2023. “Local habitat type influences bumble bee pathogens loads and bee species distributions.” *Environmental Entomology* 52:3 May 2023, paragraph 10 of the paper, paragraph 3 of the methods section, or paragraph 1 of the pathogen quantification section).

b. Test Type

The test that the researchers used was a Kruskal-Wallis test and then a Dunn post-hoc test (Gratton, E., et al., 2023. “Local habitat type influences bumble bee pathogens loads and bee species distributions.” *Environmental Entomology* 52:3 May 2023, paragraph 18 of the paper, paragraph 11 of the methods section, which is also paragraph 1 of the statistical analysis portion).

c. Hypotheses

H_O : There is no difference in the median BQCV pathogen load of *Bombus impatiens* worker in valleys, edges, and forests.

H_A : There is a difference in the median BQCV pathogen load of *Bombus impatiens* worker in valleys, edges, and forests.

d. Clean and Organizing Data

```
#create object bee_pathogens_clean
bee_pathogens_clean <- bee_pathogens |>
  # clean names, making lower case and underscores
  clean_names() |>
  # remove na values
  drop_na() |>
  # keep only site type and BQCV columns
  select(site_type_x, bqcv) |>
  # rename site type to habitat
  rename(habitat = site_type_x)
#display 10 rows from bee_pathogens_clean
slice_sample(bee_pathogens_clean, n = 10, by = NULL, weight_by = NULL,
              replace = FALSE)
```

	habitat	bqcv
1	forest	8.671499
2	valley	9.079559
3	edge	9.069267

```

4     edge  6.146976
5    valley  7.640937
6    valley  9.700377
7     edge  3.915352
8    valley 11.262078
9    forest  1.098139
10   valley  4.727993

```

e. Summarizing and Table Displays

```

# create object bee_pathogens_summary
bee_pathogens_summary <- bee_pathogens_clean |>
  # group by habitat type
  group_by(habitat) |>
  # start the summary for bee_pathogen_summary data
  summarize(
    #calculate the median BQCV load
    median_bqcv = round(median(bqcv), 1),
    # calculate the number of observations
    n = n())
# display summary calculations with flextable
flextable(bee_pathogens_summary) |>
  # remove underscore and capitalize
  set_header_labels(median_bqcv = "Median BQCV", habitat = "Habitat",
                    n = "Sample Size") |>
  # autofit to fit titles on one line
  autofit()

```

Habitat	Median BQCV	Sample Size
edge	6.3	27
forest	5.0	24
valley	7.6	29

f. Do a Test

```

# start kruskal-wallis test
kruskal.test(
  # formula: bqcv load as a function of habitat
  bqcv ~ habitat,
  # data: clean bee pathogen data frame

```

```
data = bee_pathogens_clean
)
```

Kruskal-Wallis rank sum test

data: bqcv by habitat

Kruskal-Wallis chi-squared = 9.8219, df = 2, p-value = 0.007365

g. Do a Post-Hoc Test

```
# start Dunn's Post-Hoc Test
dunn_test(
  #formula: bqcv ratings as a function of habitat
  bqcv ~ habitat,
  # data: clean bee pathogen data frame
  data = bee_pathogens_clean
)
```

A tibble: 3 x 9

	.y.	group1	group2	n1	n2	statistic	p	p.adj	p.adj.signif
*	<chr>	<chr>	<chr>	<int>	<int>	<dbl>	<dbl>	<dbl>	<chr>
1	bqcv	edge	forest	27	24	-1.83	0.0679	0.136	ns
2	bqcv	edge	valley	27	29	1.31	0.189	0.189	ns
3	bqcv	forest	valley	24	29	3.13	0.00175	0.00525	**

h. Calculate an effect size

```
# start effect size calculation
kruskal_effsize(
  # formula: bqcv ratings as a function of habitat
  bqcv ~ habitat,
  # data: clean bee pathogen data frame
  data = bee_pathogens_clean
)
```

A tibble: 1 x 5

	.y.	n	effsize	method	magnitude
*	<chr>	<int>	<dbl>	<chr>	<ord>
1	bqcv	80	0.102	eta2[H]	moderate

i. Written Communication

There was a moderate effect ($\eta^2 = 0.102$) of habitat type on BQCV loads (**Fig. 2a. Kruskal-Wallis, $\chi^2 = 9.82$, $df = 2$, $P = 0.007$, $\alpha = 0.05$**) with significant differences between the median values of valley (**$n = 29$, $\bar{x} = 7.6$**) and forest (**$n = 24$, $\bar{x} = 5.0$, $\chi^2 = 9.82$, Dunn's Test, $z = -3.13$, $P\text{-adj} = 0.005$**). While edge habitats had a greater median (**$n = 27$, $\bar{x} = 6.3$, $\chi^2 = 9.82$**) than forest there was no difference found (Dunn's Test, $z = -1.83$, $P\text{-adj} = 0.236$), and while edge habitat median was less than valley median, no difference was found (Dunn's Test, $z = 3.13$, $P\text{-adj} = 0.189$)